

Sem-GeoDF: Adaptive Gated Union of Semantic and Geometric Dynamic Feature Rejection for Stereo-Inertial Visual Odometry

ANONYMOUS AUTHOR¹

¹Affiliation withheld pending author list finalization (IEEE Access submission draft).

Corresponding author: Anonymous Author (e-mail: anonymous@example.com).

Manuscript prepared with the IEEE Access `ieeeaccess` class. Code and configs will be released upon acceptance.

ABSTRACT Dynamic objects break the static-scene assumption used by feature-based stereo-inertial visual odometry (VIO). Semantic segmentation masks can suppress movers but often over-reject useful structure in static scenes, whereas geometry-only epipolar filters need no training labels yet may fail when movers dominate the image. This paper presents **Sem-GeoDF**, an engineering fusion module for stereo-inertial VINS-Fusion that combines a non-blocking YOLO dynamic mask with GeoDF-Adaptive epipolar rejection through an adaptive gated union (logical OR), not intersection. Each branch keeps its own scene gate; candidates are merged under a shared reject-ratio budget; and a three-state online policy (static-safe / dynamic-assist / strong-dynamic) modulates soft masking, hard cull, and adaptive backend residual weights $w_i \in [w_{\min}, 1]$. The policy uses only online signals—burst ratio, strong semantic EMA, and a Dice agreement score with an explicit minimum-support requirement on raw GeoDF candidates—and never reads dataset labels, ground truth, or hold-out metrics. Hard rejection additionally requires healthy experts, observable camera motion and sufficient feature redundancy, so deletion is gated by a per-track lifecycle rather than by evidence alone. Under a fair $1.0\times$ bag-rate protocol with thresholds tuned on VIODE `city_day` only, Sem-GeoDF improves high-dynamic hold-out scenes versus GeoDF-Adaptive (e.g., `city_night/3_high` -56.9%, `parking_lot/3_high` -38.2%) while reducing static-scene regression relative to always-on semantic masking (SAD-Sem). We do not claim per-cell superiority over SAD-Sem on mean ATE; the contribution is spectrum robustness, protocol hygiene, and a reproducible evaluation on VIODE ($N = 3$ trials per cell) and EuRoC as a static-scene safety check ($N = 1\text{--}2$ trials per cell). The numerical results reported here predate the correctness and fairness fixes described in Sec. VI and are being regenerated; the method description and its test-enforced correspondence to the implementation are current. On the eight dynamic hold-out scenes, Sem-GeoDF is better than GeoDF-Adaptive on 6 and worse on 2 under a $\pm 3\%$ band.

INDEX TERMS Dynamic feature rejection, stereo-inertial odometry, semantic segmentation, sensor fusion, visual-inertial SLAM, residual weighting, YOLO

I. INTRODUCTION

STEREO-inertial visual odometry (VIO) estimates ego-motion by fusing camera tracks with inertial measurements and is a core building block for mobile robots, drones, and AR navigation [1], [2]. Most feature-based front-ends assume that the majority of tracked points lie on a rigid static scene. That assumption fails in traffic-like environments: moving vehicles inject inconsistent image motion into the visual residual and inflate absolute trajectory error (ATE), as

stressed by dynamic benchmarks such as VIODE [3].

Two complementary remedy families are widely used. Semantic methods remove features that fall on detector or instance/segmentation masks [4]–[6]. They are effective when the model fires correctly, but false positives can strip static structure, and accuracy becomes coupled to model domain, class taxonomy, and compute budget. Geometric methods reject tracks that violate multi-view rigidity, typically via epipolar Sampson tests and robust fundamental-matrix es-

timation [7]. They require no semantic training data, yet a fundamental matrix estimated from a mover-dominated image can itself become unreliable.

This paper addresses a practical systems question for stereo-inertial VINS: How can semantic and geometric experts be combined online, without oracle scene labels, so that static safety stays close to geometry-only behavior while high-dynamic accuracy exceeds geometry alone? We propose **Sem-GeoDF**, a policy-gated union (OR) of YOLO and GeoDF-Adaptive candidates under one shared reject-ratio guard, with optional adaptive backend residual weighting for suspicious survivors.

a: Contributions.

- 1) An online adaptive gated-union front-end for stereo-inertial VINS that ORs semantic and GeoDF dynamic candidates with a shared ratio/min-feature guard and non-blocking mask synchronization.
- 2) A label-free three-state policy (static-safe / dynamic-assist / strong-dynamic) driven by burst/strong EMA and a support-guarded Dice agreement score on raw GeoDF candidates, with holds expressed as durations on the sensor clock rather than frame counts.
- 3) Adaptive backend residual weights w_i with Ceres $\sqrt{w_i}$ scaling, computed only from online semantic/geometric evidence and per-id recovery.
- 4) A reproducible evaluation: fair $1.0\times$ bag rate, thresholds selected on `city_day` only, hold-out `city_night/parking_lot` ($N = 3$) and an Eu-RoC static check ($N = 1-2$ trials per cell), with a four-method matrix switched by METHODS. All reported tables are regenerated directly from run artifacts by one script, with the per-cell trial count printed in every cell.
- 5) A correctness and fairness layer that makes the evaluation checkable rather than asserted: all methods are generated from one common backbone with a machine-checked ownership allowlist; a stereo physical validity contract is shared by the backend and the geometric branch; the fundamental matrix is tested for degeneracy before it may authorise a deletion; failure detection is structured and recorded; and no publication asset can be produced from a run tree that has not passed an expected-matrix gate.
- 6) Honest positioning: we emphasize spectrum robustness and static safety rather than claiming universal best ATE against semantic-only baselines, and we state explicitly which evidence is still outstanding (Sec. VI).

II. RELATED WORK

A. DYNAMIC VISUAL AND VISUAL-INERTIAL SLAM

Handling dynamic scenes has been a long-standing goal for feature-based SLAM [8]. DynaSLAM and DS-SLAM combine geometric motion cues with semantic segmentation for visual SLAM [4], [5], and DynaSLAM II tightly couples multi-object tracking with the estimator [9]. More recent semantic-geometric systems refine this idea: RDS-SLAM re-

TABLE 1. Conceptual design contrast among fusion styles.

Aspect	AND	Sequential	Sem-GeoDF
Reject if either expert fires	no	partial	yes (OR)
Shared reject-ratio budget	rare	no	yes
Static-safe semantic hard-cull off	rare	no	yes
Label-free online policy	varies	varies	yes
Adaptive backend residual weights	rare	rare	yes (optional)

moves the tracking-thread stall of semantic masks [10], CFP-SLAM uses a coarse-to-fine dynamic-probability model [11], Crowd-SLAM targets crowded scenes with detection [12], and DytanVO jointly refines odometry and motion segmentation [13]. On the visual-inertial side, DynaVINS adds robust pose-graph factors for dynamic environments [6], and Liu et al. [14] combine object detection with IMU for resource-restricted RGB-D robots. Those full-stack systems are strong references and generally target either RGB-D or full SLAM back-ends. Our scope is intentionally narrower and complementary: a drop-in stereo-inertial front-end module for VINS-Fusion with non-blocking mask synchronisation, an online label-free policy, optional backend residual weighting, and protocol hygiene (no label leakage into the online estimator). We describe the design as non-blocking rather than real-time: a real-time claim requires end-to-end mask age and frame-drop rate to be measured and reported, which we do in Sec. V-G rather than assert. We do not claim to outperform these full stacks; we target a reproducible, honestly-scoped fusion component.

B. GEOMETRY-ONLY DYNAMIC FEATURE REJECTION

Epipolar geometry and Sampson residuals are classical tools [7]. Always-on geometric filters risk removing static structure during aggressive rotation or low parallax. GeoDF-Adaptive introduces scene-aware activation with an auto-calibrated ρ_{on} and track-level voting; Sem-GeoDF reuses this branch unchanged as the geometric expert.

C. SEMANTIC MASKING IN VIO

Real-time instance/segmentation networks such as YOLO [15], [16] provide dynamic-object pixels at interactive rates. Our SAD-Sem ablation applies semantic masking alone under the same fair bag rate and therefore isolates the benefit of fusion versus “semantics always on.”

D. HOW SEM-GEODF DIFFERS

Table 1 and Fig. 1 contrast design choices. AND fusion rejects only when both experts agree and can miss movers when one expert is silent. Sequential cascades apply one filter after the other without a shared budget. Sem-GeoDF uses policy-gated OR with a shared reject-ratio guard and a static-safe mode that disables semantic hard cull until online evidence escalates.

III. METHOD

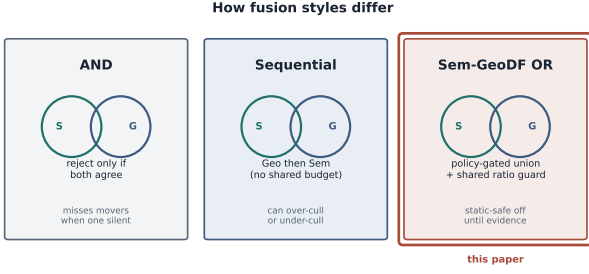


FIGURE 1. Conceptual contrast of fusion styles. Sem-GeoDF uses a policy-gated union (OR) with a shared reject budget rather than AND agreement or an unbudgeted sequential fusion.

A. SYSTEM OVERVIEW

Fig. 2 gives a full-system view of Sem-GeoDF inside stereo-inertial VINS-Fusion; Fig. 3 compresses the same front-end into a per-frame flow. Camera images feed YOLO asynchronously (`sem_block_on_mask=0`). If the latest mask is older than `sem_mask_max_age_ms`, the semantic branch is disabled for that frame (fail-open to geometry). Previous tracks feed GeoDF-Adaptive. Confirmed semantic and geometric candidates are unioned, clipped by `geodf_max_reject_ratio` and a minimum feature count, then optionally enter backend optimization with residual weights w_i .

a: Design principles.

- No oracle leakage: the estimator never reads VIODE level, ground truth, ATE/RPE, or summary files.
- Fail-open semantics: missing/stale masks never stall VINS.
- Union under a budget: OR fusion cannot wipe the track set.
- Train/hold-out hygiene: policy thresholds are frozen from `city_day` before inspecting hold-out ATE.

B. GEOMETRIC BRANCH (GEODF-ADAPTIVE)

A temporal fundamental matrix is estimated from track correspondences. RANSAC inliers/outliers and Sampson residuals form a dual gate; tracks accumulate votes over `geodf_vote_frames`. Scene activation uses an EMA of the epipolar outlier ratio with hysteresis around an auto-calibrated ρ_{on} . Confirmed geometric candidates enter the shared reject set whenever GeoDF is armed.

C. SEMANTIC BRANCH

YOLO produces a dynamic-class mask on cam0. Per-track hits are vote-confirmed over `sem_vote_frames`. A semantic scene gate tracks an EMA of the dynamic pixel ratio with hysteresis. Soft masking of new features and hard cull of existing tracks are policy-dependent (Sec. III-E).

D. EXPERT AGREEMENT WITH MINIMUM SUPPORT

An earlier implementation computed overlap only on GeoDF confirmed tracks while `frame_active`, which empirically

produced 0% overlap triggers even when both branches fired. Sem-GeoDF therefore measures agreement on the raw GeoDF candidate pool G (fallback to confirmed if empty) and the semantic-raw set S (Fig. 4).

A directed maximum, $o = \max(|G \cap S|/|G|, |S \cap G|/|S|)$, was used initially, but it is not a safe agreement measure: two two-element sets sharing a single track score $o=0.5$ and clear a 0.35 threshold on coincidence alone. We use the symmetric Sørensen–Dice coefficient

$$o_D = \frac{2|G \cap S|}{|G| + |S|} \quad (1)$$

which cannot be dominated by the smaller set, together with an explicit minimum support requirement

$$|G| \geq n_G, \quad |S| \geq n_S, \quad |G \cap S| \geq n_I. \quad (2)$$

When (2) is not met the agreement signal is set to zero rather than passed to a threshold comparison, because a ratio computed from two or three tracks is not evidence. Optionally the score is scaled by a support confidence $o_{sup} = o_D \cdot \min(1, |G \cap S|/n_{sat})$, which downweights small but qualifying intersections. The chosen metric is EMA-smoothed with coefficient `sem_policy_overlap_ema`. Directional max, directional min, Dice, Jaccard and Dice-with-support are all implemented and selected by `sem_policy_overlap_metric`, so the choice is an ablation rather than an assumption (Table 8).

E. THREE-STATE ADAPTIVE POLICY

Let the discrete state be $s \in \{0, 1, 2\}$ (static-safe, dynamic-assist, strong-dynamic), as summarized in Fig. 5. Write r_t for the current-frame dynamic-pixel ratio, $\bar{r}_t$ for its EMA, and $\bar{o}_t$ for the overlap EMA of Sec. III-D. Escalation uses three online predicates:

$$\text{burst} = [r_t \geq \tau_{burst}], \quad (3)$$

$$\text{strong} = [\bar{r}_t \geq \tau_{strong}], \quad (4)$$

$$\text{overlap} = [\bar{o}_t \geq \tau_{ov} \wedge (2)], \quad (5)$$

with defaults / train-selected values in Table 2.

a: The hold is a duration, not a frame count

The hold was originally H frames (`sem_policy_hold_frames=`). A frame count is not a well-defined quantity: at 10 Hz it is 18 s, at 30 Hz it is 6 s, and it shortens further whenever frames are dropped, so the same configuration means different things on different sensors. The hold is now a duration on the sensor clock (never wall-clock),

$$t^{\text{hold}} \leftarrow \max(t^{\text{hold}}, t + \Delta_{\text{assist}}), \quad \text{hold active} = [t < t^{\text{hold}}], \quad (6)$$

with separate Δ_{assist} and Δ_{strong} . A minimum dwell time prevents state chatter, and blocks only downgrades: escalation is never delayed, because waiting is the unsafe direction when real dynamic content appears. A backwards timestamp (bag restart) resets the machine rather than leaving a hold armed for the remainder of the run.

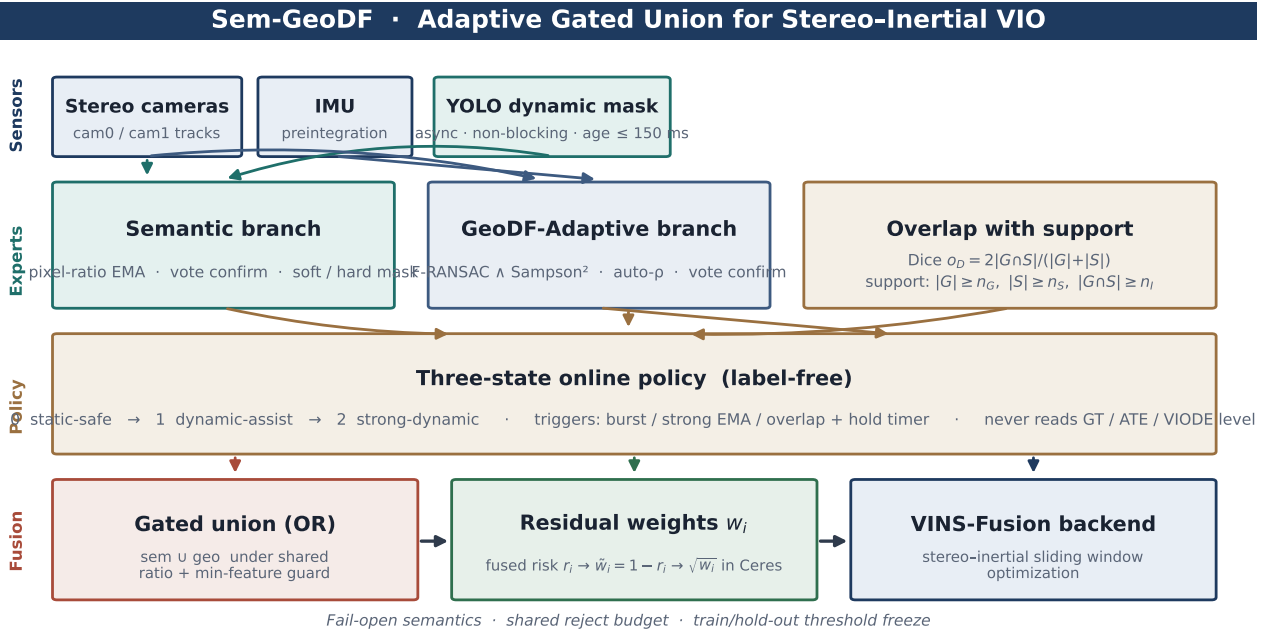


FIGURE 2. Sem-GeoDF system overview. Sensors feed a semantic expert (YOLO) and a GeoDF-Adaptive geometric expert; bidirectional overlap and a label-free three-state policy.

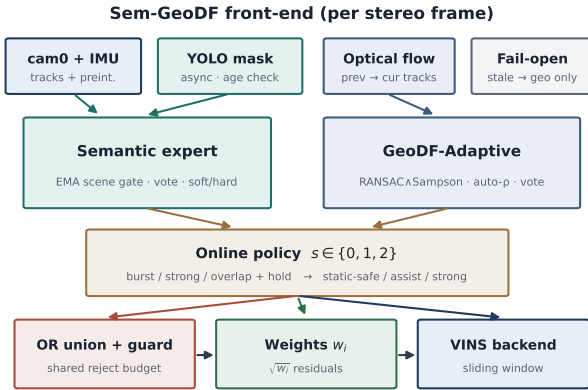


FIGURE 3. Per-frame Sem-GeoDF front-end: fail-open mask sync, dual experts, online policy, gated OR under a shared reject budget, and $\sqrt{\tilde{w}_i}$ residual weighting into VINS.

- $s=0$ (static-safe): soft-mask new features when the semantic scene is active; semantic hard cull off.
- $s=1$ (dynamic-assist): entered on burst, overlap, or an active hold; vote-confirmed semantic tracks enter the shared reject set when the semantic scene is active.
- $s=2$ (strong-dynamic): entered on strong EMA and/or assist plus usable raw geo evidence; full OR fusion armed under the same shared reject budget.

Final selected values are written to `selected_params.yaml` from `city_day` replay only (never from hold-out ATE).

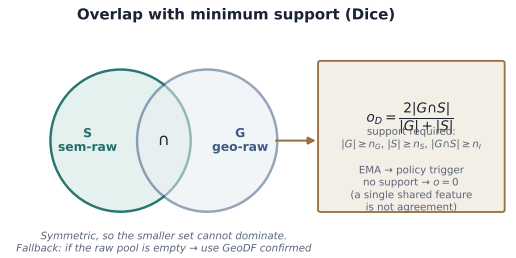


FIGURE 4. Expert agreement on the raw GeoDF pool. Dice is symmetric, so the smaller set cannot dominate.

TABLE 2. Key Sem-GeoDF parameters (defaults / train-selected).

Parameter	Value
sem_vote_frames / geodf_vote_frames	2 / 2
sem_policy_burst_ratio	0.18
sem_policy_strong_ratio	0.18 (selected)
sem_policy_hold_frames	180 (selected)
sem_policy_overlap_ratio	0.35
geodf_max_reject_ratio	0.40
sem_mask_max_age_ms	150
sem_geodf_backend_weight	1 (enabled)
sem_geodf_backend_min_weight	0.25
sem_geodf_backend_recovery	0.20

F. HEALTH, OBSERVABILITY AND PER-TRACK LIFECYCLE

Neither expert is trustworthy unconditionally, so evidence is not allowed to act directly. Three health scores in $[0, 1]$ are measured first, and only then does a policy decide what may

Three-state adaptive policy (online, label-free)

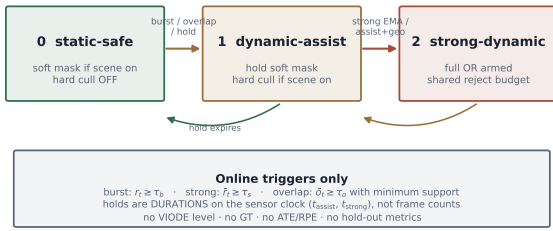


FIGURE 5. Three-state online policy. Escalation uses burst, strong semantic EMA, and bidirectional overlap-only; the estimator never reads VIODE level, ground truth, or trajectory.

be done.

Semantic health h_{sem} falls with mask staleness (using the mask's own source timestamp, Sec. III-C) and with mask saturation: a mask that labels most of the frame dynamic is far more often a segmentation failure than a scene in which everything moves, so it must not be permitted to delete features.

Geometric health h_{geo} comes from an explicit degeneracy test on the fundamental matrix, rather than from F merely being non-empty. F estimated during near-pure rotation, at low parallax, from correspondences clustered in one image region, from weak RANSAC support, or from a mover-dominated frame is not empty — it is confidently wrong, and hard-rejecting static structure on it is precisely the failure mode this work claims to avoid. We therefore require a minimum median parallax, a minimum grid occupancy, a bounded residual-spread condition number, and minimum RANSAC inlier count and ratio. Low parallax and an ill-conditioned estimate mark the geometry degenerate (no GeoDF decision at all); clustered features, weak support and mover dominance mark it weak (down-weight only, and not admissible as strong two-expert agreement).

Observability health h_{obs} is the minimum of normalized redundancy (track count), median parallax, and spatial spread. The minimum, not a mean: many features clustered in one corner do not make camera motion observable.

Each track then carries its own lifecycle,

Trusted $\rightarrow$ Suspect $\rightarrow$ DownWeighted $\rightarrow$ Rejected,

with Suspect/DownWeighted $\rightarrow$ Recovering $\rightarrow$ Trusted when evidence disappears and the track stays clean for a dwell time. Hard rejection is the only irreversible action, and a track may be deleted only when all of the following hold:

$$r_i \geq \tau_{\text{rej}} \wedge \text{two-expert agreement} \wedge h_{\text{sem}}, h_{\text{geo}} \text{ healthy} \wedge h_{\text{obs}} \text{ healthy} \wedge \text{track is finished} \quad (7)$$

and a per-frame deletion allowance of $N_{\text{track}} - N_{\text{min}}$ bounds how many tracks a single frame may remove. Without that allowance, evaluating $N_{\text{track}} \geq N_{\text{min}}$ once per track lets a batch of simultaneous deletions overshoot the floor. Anything short of (7) is a recoverable down-weight, and which guard

blocked a deletion is logged so the behaviour is auditable rather than invisible.

This is the substantive difference from an OR fusion under a reject-ratio cap: what the method manages is a factor lifecycle governed by evidence quality, redundancy and estimator observability.

G. STEREO PHYSICAL VALIDITY CONTRACT

A stereo match previously had to satisfy only optical flow plus a fixed left–right cycle tolerance. Nothing verified that it was physically possible, so a match with the wrong disparity direction or a negative triangulated depth could reach the stereo factor, depth initialisation, the right-camera geometric branch and the weight evidence.

Every left–right match now passes one contract before any consumer sees it: flow status and border validity; a left $\rightarrow$ right $\rightarrow$ left cycle bound; an epipolar residual against $E = [t]_{\times} R$ from the actual extrinsic; a disparity magnitude within $[d_{\text{min}}, d_{\text{max}}]$; positive triangulated depth in both cameras; and a bounded distance between the two rays at closest approach.

Crucially, the expected disparity direction is derived per feature rather than hard-coded. For a ray $\mathbf{x}_0$ in cam0 at inverse depth $w = 1/Z$, the cam1 projection is $\pi(R\mathbf{x}_0 + w\mathbf{t})$, which starts at the point at infinity $\pi(R\mathbf{x}_0)$ for $w=0$ and moves along the epipolar line as w grows; a physically possible match lies on that side. This is exact for any rig geometry, so the same test is valid for left–right, right–left, vertical and verged rigs, whereas “disparity must be positive” is a calibration-specific assumption.

There is exactly one stereo matcher. The geometric branch reads only contract-validated observations instead of running a second optical flow of its own, which previously let its measurement of a track disagree with the measurement the backend factor used for the same track. Per-cause counters (wrong disparity sign, cycle, epipolar, negative depth, re-projection, range, border) are logged for every frame, so a published rejection rate is accountable.

H. FAILURE DETECTION

A diverged run must be reported as failed, not emit a plausible-looking trajectory that then enters a table. The estimator evaluates, per frame and most-severe-first: non-finite state, solver failure, accelerometer- and gyroscope-bias divergence, translation and rotation jumps, and a sustained run of low-feature frames (a single sparse frame is normal). The reason is structured (NanState, SolverFailure, AccBiasDivergence, GyroBiasDivergence, TranslationJump, RotationJump, LowFeatureCount) and written to the run manifest with its timestamp, so failures are counted rather than inferred.

I. ADAPTIVE BACKEND RESIDUAL WEIGHTING

Hard rejection is not always possible: a shared ratio guard and minimum-feature constraints may keep suspicious tracks alive. When `sem_geodf_backend_weight=1`, each

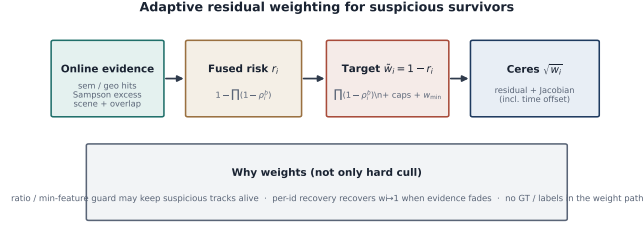


FIGURE 6. Adaptive residual-weight path for tracks that survive the shared reject budget. Apply semantic branch / min-feature-guard to R_i ; soft-masking (noisy-OR) allowed

surviving track i carries a residual weight $w_i \in [w_{\min}, 1]$, broadcast to all of its observations and combined as the per-pair minimum across the two frames of a factor (Fig. 6). Every visual factor in Ceres multiplies both the residual and all of its Jacobian blocks (including the time-offset term) by $\sqrt{w_i}$, so the contribution is a genuine weighted least-squares term rather than an untracked post-hoc score. The weight is thus a per-track confidence, not an independent per-observation noise model; we state this explicitly to avoid overclaiming.

From online evidence at the current frame we form three risk terms in $[0, 1]$:

$$\rho_i^{\text{sem}} = \mathcal{K}_i^{\text{sem}} (1 - w_{\text{sem}}) c_i^{\text{sem}}, \quad (8)$$

$$\rho_i^{\text{geo}} = \mathcal{K}_i^{\text{geo}} (1 - w_{\text{geo}}) \max(c_i^{\text{geo}}, e_i), \quad (9)$$

$$\rho_i^{\text{agr}} = \mathcal{K}_i^{\text{sem}} \mathcal{K}_i^{\text{geo}} (1 - w_{\text{agr}}) \max(\bar{o}_t, \min(c_i^{\text{sem}}, \max(c_i^{\text{geo}}, e_i))), \quad (10)$$

where $\mathcal{K}_i^{\text{sem}} / \mathcal{K}_i^{\text{geo}}$ flag semantic / GeoDF raw-or-confirmed hits, $c_i^{\text{sem}} / c_i^{\text{geo}}$ are scene confidences from EMA+hysteresis (and policy state), and e_i is the normalized Sampson excess. A multiplicative fusion (noisy-OR-inspired; we do not assume the three evidence terms are independent, so we avoid calling it a probabilistic OR) combines them into a single fused risk

$$r_i = 1 - \prod_{b \in \{\text{sem}, \text{geo}, \text{agr}\}} (1 - \rho_i^b), \quad (11)$$

which is then mapped to a target weight by inverting the risk,

$$\tilde{w}_i = \text{clip}(1 - r_i, w_{\min}, 1) = \text{clip}\left(\prod_b (1 - \rho_i^b), w_{\min}, 1\right), \quad (12)$$

so that more dynamic evidence always yields a smaller weight. Afterwards $\tilde{w}_i$ is capped by w_{sem} , w_{geo} , or w_{agr} when the corresponding confirmation flags fire, and re-clipped to $[w_{\min}, 1]$. Eqs. (11)–(12) are enforced against the implementation by the property test `PaperEquationMatchesImplementation`, together with the monotonicity property that raising any evidence confidence never raises $\tilde{w}_i$. Per-id recovery with rate α_{rec}

Algorithm 1 Sem-GeoDF front-end step (per stereo frame)

- 1: Sync latest YOLO mask if age $\leq$ max age; else disable semantic branch
- 2: Update semantic EMA; run GeoDF-Adaptive $\rightarrow$ raw/confirmed sets
- 3: Compute bidirectional overlap o on raw GeoDF pool; update policy s
- 4: $R \leftarrow$ GeoDF confirmed $\cup$ (policy-gated semantic confirmed)
- 5: Apply semantic branch / min-feature-guard to R_i ; soft-masking (noisy-OR) allowed
- 6: If backend weighting enabled, assign w_i for survivors and recover idle ids

raises w_i toward a higher target gradually so transient false positives do not permanently suppress a track:

$$w_i \leftarrow \begin{cases} w_i^{\text{prev}} + \alpha_{\text{rec}}(\tilde{w}_i - w_i^{\text{prev}}) & \text{if } \tilde{w}_i > w_i^{\text{prev}}, \\ \tilde{w}_i & \text{otherwise.} \end{cases} \quad (13)$$

a: Consistency with marginalization

Each observation's weight is frozen when its residual block is constructed ($\sqrt{w_i}$ is a const member of the factor), so:

Recovery changes the weight of new observations of the same track; residuals that have already been marginalized retain the weight they were created with.

The marginalization factor therefore never reads a mutable global weight, and updating a track's current weight cannot retroactively alter a prior that has already been formed. This invariant is checked by `test_marginalization_weight_freeze`.

Audit columns in `sem_geodf_stats.csv` separate the three quantities of Sec. III-I: `mean_target_weight_pre_guard` (mapping output), `mean_applied_weight_post_guard` and `min_survivor_weight` (after recovery, over tracks that survived the shared reject budget), plus `weighted_candidates_pre_guard`, `weighted_survivor` and `rejected_weighted_tracks` so that down-weighting is reported for survivors rather than for candidates. Table 7 lists the defaults used in paper configs.

IV. EXPERIMENTAL SETUP

A. IMPLEMENTATION

We use a ROS 2 Humble stereo-inertial VINS derived from VINS-Fusion [2]. YOLO runs on CUDA. All methods share a fair bag playback rate of $1.0\times$ (including non-YOLO baselines), so semantic methods are not artificially slowed relative to geometry-only runs.

B. DATASETS

VIODE [3]: environments `city_day`, `city_night`, and `parking_lot`, each with dynamic levels `0_none`, `1_low`, `2_mid`, `3_high`. **EuRoC** [17]: Machine Hall sequences `MH_01`–`MH_05`, used as a static-scene safety

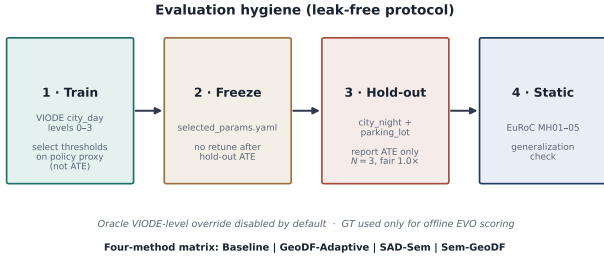


FIGURE 7. Leak-free evaluation protocol: thresholds are selected on VIODE city_day only (policy proxy, not ATE), frozen, then reported on hold-out night/parking and EuRoC.

check. We do not present this as evidence of generalization: VIODE is simulated and EuRoC is static, so neither speaks to real-world dynamic scenes. A generalization claim would need at least one real dynamic sequence, which we list as required future evidence in Sec. VI.

C. METHODS

Four methods are compared via METHODS:

- **Baseline:** stereo-inertial VINS without dynamic rejection.
- **GeoDF-Adaptive:** geometry-only adaptive rejection.
- **SAD-Sem:** semantic-only YOLO masking.
- **Sem-GeoDF (ours):** adaptive gated union + backend weights.

Optional sequential / mask-gated ablations exist in code but are excluded from the default matrix to keep the comparison focused.

D. PROTOCOL

Policy thresholds are selected on `city_day` only by maximizing a weighted static/dynamic policy objective on logged stats (not ATE). Hold-out night/parking ATE and EuRoC are reported after freezing thresholds (Fig. 7). Primary metric: ATE RMSE after SE(3) Umeyama alignment (EVO). On VIODE we run $N = 3$ trials per cell after an $N=1$ gate used to catch diverged runs; EuRoC uses $N = 1-2$ trials per cell (fewer than VIODE), so we treat it as a static generalization check rather than a full variance estimate. Every table cell prints its own trial count as a superscript, so $N=1$ cells are never mistaken for averaged means. All tables, figures, and abstract numbers are emitted by a single script (`make_sem_geodf_paper_assets.py`) directly from the run artifacts, so the manuscript is reproducible from the result tree.

V. RESULTS

A. VIODE TRAJECTORY ACCURACY

Fig. 8 summarizes Sem-GeoDF versus GeoDF-Adaptive on VIODE (negative = better). The largest gains concentrate on high-dynamic hold-out night and parking scenes, where geometry alone is insufficient. As Table 4 separates, on the eight dynamic hold-out cells Sem-GeoDF is better on 6,

worse on 2, with the two regressions at the low-dynamic `0_none/1_low` end where semantics add little. We therefore claim spectrum consistency on hard scenes, not universal per-cell dominance.

B. EUROC STATIC-SCENE BEHAVIOUR

Fig. 9 highlights selected static and high-dynamic scenes across the four methods.

C. INTERPRETATION VERSUS SAD-SEM

The large negative deltas in Table 6 are measured against GeoDF-Adaptive, i.e. the geometry-only branch, which is the weaker baseline on high-dynamic night/parking scenes. Against semantic-only SAD-Sem the picture is different: on those same hard cells SAD-Sem often achieves the best per-cell ATE (e.g. `city_night_3_high` SAD-Sem ≈ 0.273 vs ours ≈ 0.354), and the mean ATE of Sem-GeoDF is typically close to SAD-Sem rather than strictly better. We therefore state the headline explicitly as gains over geometry-only filtering, and we do not market Sem-GeoDF as a strict ATE winner over SAD-Sem. Its value for IEEE Access is the combination of: (i) large gains over geometry-only filtering on hard hold-outs; (ii) reduced static regression relative to always-on semantics (Sec. V-D); (iii) label-free online control, shared reject budget, and adaptive residual weights; (iv) a reproducible fair-rate protocol whose tables regenerate exactly from the run artifacts.

D. STATIC SAFETY

On `0_none` levels, always-on semantics can regress relative to Baseline because false-positive masks remove useful structure. Table 6 quantifies the effect: on `city_day_0_none`, SAD-Sem is +36% versus Baseline while Sem-GeoDF stays within +5%; on `parking_lot_0_none`, SAD-Sem is +12% while Sem-GeoDF is -17% (best of the four). `city_night_0_none` is mixed—both Adaptive and SAD-Sem improve over Baseline, and Sem-GeoDF sits between them—but the static-safe policy still avoids the day/parking over-rejection pattern of always-on masking. This is the practical justification for a policy rather than always-on semantics.

E. ROLE OF ADAPTIVE RESIDUAL WEIGHTS

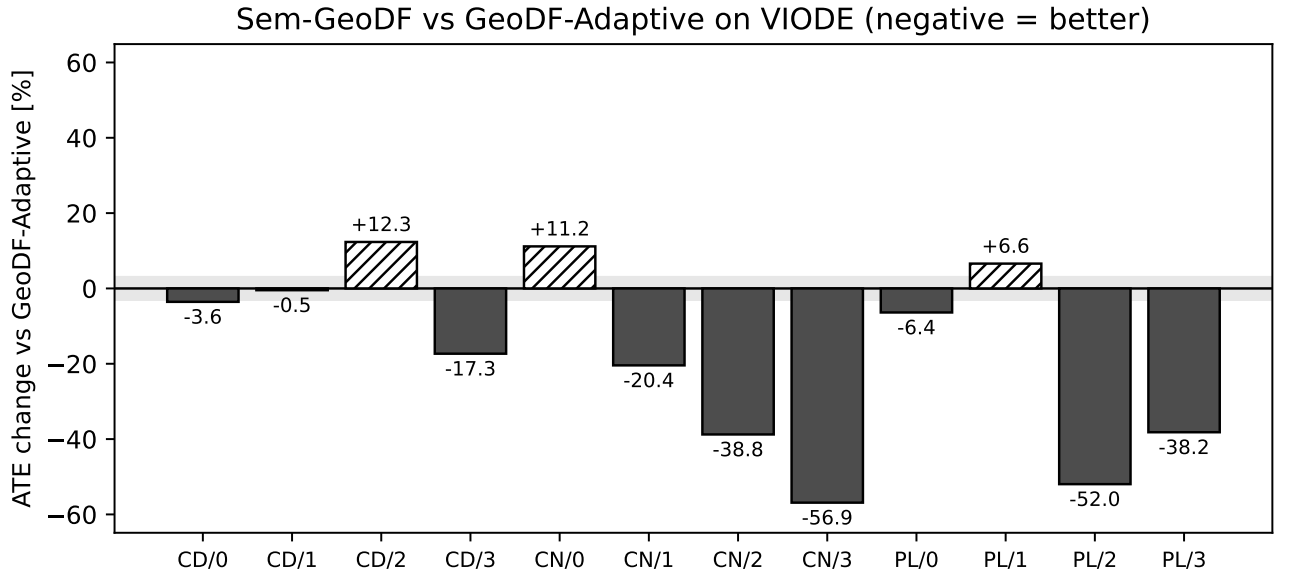
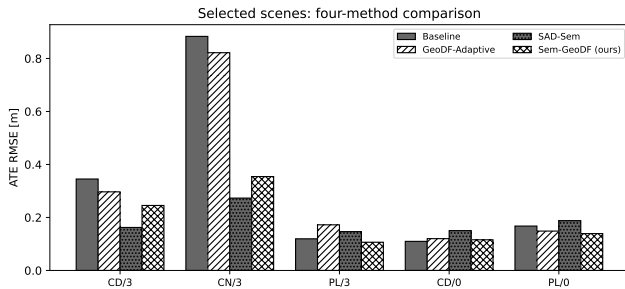
Backend weighting complements hard cull: when the ratio guard preserves suspicious tracks, down-weighting reduces their influence without emptying the feature set. Because weights are recovered per id, transient false positives do not permanently suppress a track. This mechanism is active in Sem-GeoDF paper configs (`sem_geodf_backend_weight=1`) and is absent from Baseline / Adaptive / SAD-Sem.

F. BACKEND WEIGHT MECHANISM

Table 7 summarizes the residual-weight parameters used throughout the Sem-GeoDF paper runs. A paired ATE ablation that disables only w_i (`sem_geodf_backend_weight=0`)

TABLE 3. VIODE ATE RMSE [m] (mean $\pm$ std). Superscript is the number of trials N per cell; cells with $N=1$ report a single run (no std). Best mean per row in bold.

Scene	Baseline	GeoDF-Adaptive	SAD-Sem	Sem-GeoDF (ours)
city_day_0_none	0.110$\pm$0.000³	0.120 $\pm$ 0.000 ³	0.150 $\pm$ 0.012 ³	0.116 $\pm$ 0.000 ³
city_day_1_low	0.139 $\pm$ 0.001 ³	0.155 $\pm$ 0.000 ³	0.137$\pm$0.000³	0.154 $\pm$ 0.002 ³
city_day_2_mid	0.166 $\pm$ 0.000 ³	0.142$\pm$0.000³	0.148 $\pm$ 0.001 ³	0.159 $\pm$ 0.000 ³
city_day_3_high	0.345 $\pm$ 0.002 ³	0.297 $\pm$ 0.020 ³	0.162$\pm$0.019³	0.245 $\pm$ 0.020 ³
city_night_0_none	0.418 $\pm$ 0.000 ³	0.307 $\pm$ 0.107 ³	0.298$\pm$0.005³	0.342 $\pm$ 0.000 ³
city_night_1_low	0.500 $\pm$ 0.000 ³	0.568 $\pm$ 0.000 ³	0.357$\pm$0.073³	0.452 $\pm$ 0.000 ³
city_night_2_mid	0.497 $\pm$ 0.000 ³	0.460 $\pm$ 0.000 ³	0.372 $\pm$ 0.000 ³	0.281$\pm$0.013³
city_night_3_high	0.883 $\pm$ 0.001 ³	0.822 $\pm$ 0.000 ³	0.273$\pm$0.030³	0.354 $\pm$ 0.000 ³
parking_lot_0_none	0.168 $\pm$ 0.001 ³	0.149 $\pm$ 0.000 ³	0.188 $\pm$ 0.000 ³	0.139$\pm$0.000³
parking_lot_1_low	0.107 $\pm$ 0.018 ³	0.106$\pm$0.000³	0.132 $\pm$ 0.000 ³	0.113 $\pm$ 0.006 ³
parking_lot_2_mid	0.144 $\pm$ 0.000 ³	0.205 $\pm$ 0.000 ³	0.088$\pm$0.000³	0.099 $\pm$ 0.000 ³
parking_lot_3_high	0.119 $\pm$ 0.000 ³	0.172 $\pm$ 0.000 ³	0.146 $\pm$ 0.003 ³	0.106$\pm$0.037³

**FIGURE 8.** ATE change of Sem-GeoDF versus GeoDF-Adaptive on VIODE (negative = better). Shaded band: $\pm 3\%$.**FIGURE 9.** Selected static and high-dynamic scenes: four-method ATE comparison.

while freezing the identical gated union and train-selected policy is reserved for camera-ready once the `sem_geodf_noweight` matrix finishes; until then we do not invent numerical deltas. The equations in Sec. III-I

already make the contribution of w_i explicit relative to hard cull under the shared reject budget.

G. SENSITIVITY AND RUNTIME BEHAVIOUR

Offline replay on `city_day` shows that nearby (burst, strong, ho grid points yield similar weighted policy objectives. Overlap thresholds 0.30–0.40 change little when overlap events are sparse—consistent with burst/strong dominating escalation on many sequences. Non-blocking mask synchronisation means the estimator never waits on the segmentation network: it either consumes a mask that is fresh by its own source timestamp or falls back to GeoDF-only. That is a statement about the coupling, not a real-time guarantee. A real-time claim additionally requires end-to-end mask age, mask reuse and drop ratios, processed-frame ratio, and per-frame estimator latency, all of which the node now publishes (`/dynamic_mask/source_stamp`, `/inference_finish_stamp`, `/reused`, `/model_latency`).

TABLE 4. Sem-GeoDF ATE change versus GeoDF-Adaptive (negative = better; both means over the same N). Decision band $\pm 3\%$. Train (*city_day*) is separated from hold-out; EuRoC is a static generalization check.

Scene	Δ vs Adaptive [%]	Verdict	N
<i>Train: city_day (thresholds selected here)</i>			
city_day_0_none	-3.6	better	3
city_day_1_low	-0.5	similar	3
city_day_2_mid	+12.3	worse	3
city_day_3_high	-17.3	better	3
subtotal: 2 better / 1 similar / 1 worse			
<i>Hold-out: city_night, parking_lot</i>			
city_night_0_none	+11.2	worse	3
city_night_1_low	-20.4	better	3
city_night_2_mid	-38.8	better	3
city_night_3_high	-56.9	better	3
parking_lot_0_none	-6.4	better	3
parking_lot_1_low	+6.6	worse	3
parking_lot_2_mid	-52.0	better	3
parking_lot_3_high	-38.2	better	3
subtotal: 6 better / 0 similar / 2 worse			
<i>EuRoC (static generalization)</i>			
MH_01_easy	+3.7	worse	2
MH_02_easy	-8.0	better	2
MH_03_medium	-0.0	similar	1
MH_04_difficult	-2.2	similar	1
MH_05_difficult	-1.7	similar	1
subtotal: 1 better / 3 similar / 1 worse			
Hold-out headline: 6 better / 0 similar / 2 worse on the 8 dynamic hold-out scenes			

and the manifest records; the corresponding table is listed in Table 8 and is pending the re-run described in Sec. VI. Logged `sem_mask_lag_ms` provides an audit of mask age at consumption time. The $\pm 3\%$ decision band is a reporting convention (roughly the run-to-run ATE spread we observe on repeated VIODE trials), used only to label a cell better/similar/worse; it is not a significance test, and all raw deltas are shown so readers can apply their own threshold.

H. FAILURE MODES AND LIMITATIONS

- 1) Mid-dynamic day cells where GeoDF already helps and adding semantics can slightly increase ATE; two low-dynamic hold-out cells (`city_night_0_none`, `parking_lot_1_low`) also regress versus GeoDF-Adaptive.
- 2) The three policy escalation triggers (burst, strong, overlap) are all driven by semantic signals; geometry never escalates the semantic branch on its own. Sem-GeoDF is thus best described as “geometry always-on with its own gate, OR policy-gated semantics,” not a symmetric consensus.
- 3) The residual-weight mechanism is specified and enabled (Sec. III-I, Table 7), but a paired ATE ablation with w_i off is still pending camera-ready completion of the `sem_geodf_noweight` matrix; we therefore do not claim a quantified ATE split between union and weighting yet.
- 4) EuRoC uses $N = 1$ –2 trials per cell (fewer than VIODE); it is a static generalization check, not a full variance estimate. $N = 3$ on VIODE is also small for

formal significance testing.

- 5) VIODE is simulated; real-world dynamic bags remain future work.
- 6) Matched external SOTA tables (e.g., DynaVINS under identical bags/rates) are not included here.

VI. STATUS OF THE NUMERICAL RESULTS

This manuscript is a work in progress, and we state plainly which parts are settled and which are not.

a: Settled

The method description (Sec. III-D–III-I) matches the released implementation, and the correspondence is enforced mechanically rather than by inspection: the weighting equations (11)–(12) are checked against the code by a property test that re-derives them independently, the projection factors are validated by finite-difference gradient tests at $w \in \{1, 0.75, 0.25\}$, and the policy, overlap, health, lifecycle, degeneracy, stereo-contract and failure-detection components each carry unit tests plus a synthetic integration test that requires neither GPU nor dataset.

b: Not settled: every number in Sec. V must be regenerated. The tables and figures reporting ATE were produced before the corrections described in this paper, and they are not valid evidence for the current method. Specifically:

- the earlier Sem-GeoDF configuration additionally enabled per-observation visual quality weighting and adaptive IMU covariance, which the Baseline, GeoDF-Adaptive and SAD-Sem configurations did not, so no measured improvement could be attributed to Sem-GeoDF itself. All methods are now generated from one common backbone with a machine-checked ownership allowlist;
- failure detection was inert (an unconditional early return), so diverged runs produced trajectories that entered the averages;
- the publication path silently excluded any run with ATE above 50 m, removing exactly the cases the method claims to improve, and did not report success rate;
- the agreement trigger used a directed maximum with no minimum support, and the hold was a frame count rather than a duration.

Each of these changes the algorithm’s behaviour, so the entire matrix has to be re-run. The re-run is a compute task, not an open research question: the protocol, the expected matrix, the paired seeds and the validation gate are all committed, and no publication asset can be generated until `validate_experiment_matrix.py` returns pass on the new run tree.

c: Tables the final version must contain

d: Claims we do not make

We do not claim state-of-the-art dynamic VIO; we do not claim to be better than semantic-only masking per cell; we

TABLE 5. EuRoC Machine Hall ATE RMSE [m]. Superscript is the number of trials N per cell; cells with $N=1$ report a single run (no std). Best mean per row in bold.

Scene	Baseline	GeoDF-Adaptive	SAD-Sem	Sem-GeoDF (ours)
MH_01_easy	0.166±0.021 ²	0.177±0.000 ²	0.175±0.000 ²	0.183±0.000 ²
MH_02_easy	0.169±0.000 ²	0.169±0.000 ²	0.158±0.000 ²	0.155±0.000 ²
MH_03_medium	0.292 ¹	0.274 ¹	0.261 ¹	0.274 ¹
MH_04_difficult	0.447 ¹	0.441 ¹	0.440 ¹	0.432 ¹
MH_05_difficult	0.298 ¹	0.298 ¹	0.294 ¹	0.293 ¹

TABLE 6. Static-scene ATE RMSE [m] on VIODE 0_none ($N=3$). Δ vs Baseline in parentheses (negative = better).

Scene	Base.	Adapt.	SAD-Sem	Sem-GeoDF
city_day_0_none	0.110	0.120 (+9%)	0.150 (+36%)	0.116 (+5%)
city_night_0_none	0.418	0.307 (−27%)	0.298 (−29%)	0.342 (−18%)
parking_lot_0_none	0.168	0.149 (−11%)	0.188 (+12%)	0.139 (−17%)

TABLE 7. Adaptive residual-weight mechanism (enabled in Sem-GeoDF paper configs; `sem_geodf_backend_weight=1`). A paired ATE ablation with w_i off under an identical gated union is planned for camera-ready once `sem_geodf_noweight` runs finish; Baseline / Adaptive / SAD-Sem do not use this term.

Symbol / flag	Default	Role
$w_{\min}$	0.25	Floor on residual weight
w_{sem}	0.55	Cap when semantic-confirmed
w_{geo}	0.75	Cap when GeoDF-confirmed
w_{agr}	0.25	Cap when both confirmed
α_{rec}	0.20	Per-id recovery toward target
Ceres scale	$\sqrt{w_i}$	Residual and full Jacobian

do not claim probabilistically optimal fusion, since the evidence terms are not independent; we do not claim real-time operation until end-to-end latency and drop rate are reported; and we do not claim generalization from simulated-dynamic plus static-real data alone.

VII. DISCUSSION

IEEE Access is the target venue because the contribution is a reproducible applied fusion module with honest trade-offs, rather than a short letter claiming frontier novelty over full dynamic-VIO stacks. The manuscript follows the official Access two-column layout (`ieeeaccess.cls`).

a: Claim framing.

We claim spectrum robustness and static safety: large gains versus GeoDF-Adaptive on high-dynamic hold-outs (Table 4), reduced static regression versus always-on semantics (Table 6), and a label-free online policy under a fair 1.0× protocol. We do not claim universal per-cell ATE superiority over SAD-Sem, matched DynaVINS SOTA under identical bags, or a finished paired w_i ATE ablation.

b: Reproducibility.

Each run writes `run_manifest.json` (git SHA, config hash, bag rate, oracle flag) and per-frame `sem_geodf_stats.csv`. Policy thresholds can be re-

TABLE 8. Evidence required before the claims in this paper are supported. None may be omitted, and the asset generator refuses to emit any of them from an unvalidated run tree.

Evidence	Status
Common-backbone ATE / RPE	pending re-run
Success rate and trajectory coverage	pending re-run
Gated union on/off (U vs S, G)	pending re-run
Backend weight on/off (U vs U+W)	pending re-run
Overlap-metric ablation	pending re-run
Static-scene safety	pending re-run
Runtime and mask-age distribution	pending re-run
Robustness injection	pending re-run
Real-world dynamic evaluation	dataset not yet available
Config / protocol summary	committed

played offline with `simulate_sem_policy.py`. The `paper_postfix` result tree is generated with the corrected $\sqrt{w_i}$ scaling on the time-offset Jacobian (Sec. III-I), so the released estimator reproduces the reported Sem-GeoDF numbers without a residual/Jacobian mismatch. The default paper matrix is launched by

```
METHODS="baseline adaptive sad_sem sem_geodf".
```

Code, configs, and evaluation scripts will be released upon acceptance.

VIII. CONCLUSION

Sem-GeoDF couples semantic and geometric dynamic-feature rejection through an online adaptive gated union for stereo-inertial VIO, with optional adaptive backend residual weighting for suspicious survivors. With label-free control and a strict train/hold-out protocol, it improves several high-dynamic VIODE hold-outs over geometry-only filtering while mitigating semantic over-rejection in static scenes. We present this work as an engineering contribution for IEEE Access with full tables, figures, explicit limitations, and a clear path to camera-ready weight ablation.

ACKNOWLEDGMENT

The authors thank colleagues who provided dataset paths, compute support, and feedback during evaluation.

REFERENCES

- [1] T. Qin, P. Li, and S. Shen, "VINS-Mono: A robust and versatile monocular visual-inertial state estimator," *IEEE Transactions on Robotics*, vol. 34, no. 4, pp. 1004–1020, 2018.
- [2] T. Qin, S. Cao, J. Pan, and S. Shen, "A general optimization-based framework for global pose estimation with multiple sensors," in *arXiv preprint arXiv:1901.03642*, 2019.

- [3] K. Minoda, R. Yonetani et al., “VIODE: A simulated dataset to address the challenges of visual-inertial odometry in dynamic environments,” in *IEEE Robotics and Automation Letters*, 2021.
- [4] B. Bescos, J. M. Fácil, J. Civera, and J. Neira, “DynaSLAM: Tracking, mapping, and inpainting in dynamic scenes,” in *IEEE Robotics and Automation Letters*, vol. 3, no. 4, 2018, pp. 4076–4083.
- [5] C. Yu, Z. Liu, X.-J. Liu, F. Xie, Y. Yang, Q. Wei, and Q. Fei, “DS-SLAM: A semantic visual SLAM towards dynamic environments,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018, pp. 1168–1174.
- [6] S. Song, H. Lim, A. J. Lee, and H. Myung, “DynaVINS: A visual-inertial SLAM for dynamic environments,” *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 11 523–11 530, 2022.
- [7] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2003.
- [8] C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, and J. J. Leonard, “Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age,” *IEEE Transactions on Robotics*, vol. 32, no. 6, pp. 1309–1332, 2016.
- [9] B. Bescos, C. Campos, J. D. Tardós, and J. Neira, “DynaSLAM II: Tightly-coupled multi-object tracking and SLAM,” *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 5191–5198, 2021.
- [10] C. Zhang, R. Zhang, S. Jin, and X. Yi, “RDS-SLAM: Real-time dynamic SLAM using semantic segmentation methods,” *IEEE Access*, vol. 9, pp. 23 772–23 785, 2021.
- [11] X. Hu, Y. Zhang, Z. Cao, R. Ma, Y. Wu, Z. Deng, and W. Sun, “CFP-SLAM: A real-time visual SLAM based on coarse-to-fine probability in dynamic environments,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 4399–4406, 2022.
- [12] J. C. V. Soares, M. Gattass, and M. A. Meggiolaro, “Crowd-SLAM: Visual SLAM towards crowded environments using object detection,” *Journal of Intelligent & Robotic Systems*, vol. 102, no. 2, p. 50, 2021.
- [13] S. Shen, Y. Cai, W. Wang, and S. Scherer, “DytanVO: Joint refinement of visual odometry and motion segmentation in dynamic environments,” *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4048–4055, 2023.
- [14] J. Liu, X. Li, Y. Liu, and H. Chen, “RGB-D inertial odometry for a resource-restricted robot in dynamic environments,” *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 9573–9580, 2022.
- [15] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.
- [16] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLO,” <https://github.com/ultralytics/ultralytics>, 2023.
- [17] M. Burri, J. Nikolic, P. Gohl, T. Schneider, J. Rehder, S. Omari, M. W. Achtelik, and R. Siegwart, “The EuRoC micro aerial vehicle datasets,” *The International Journal of Robotics Research*, vol. 35, no. 10, pp. 1157–1163, 2016.

ANONYMOUS AUTHOR Anonymous Author is preparing this IEEE Access manuscript. Replace this placeholder with a short biography for every listed author before submission (IEEE Access requires biographies for all authors).

...